Pooja Kiran

Security Engineer | AI & Agent Security | Application Security | Cloud IAM Security | Model Supply-Chain Security

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Experience

Independent AI Security Researcher & Engineer Aug. 2024 - Present

Self-Directed Research *Tempe, AZ, USA*

- Implemented MCP inspection controls, offline IAM checks, and non-executing model-artifact analysis across three self-directed repositories.
- Automated pytest and GitHub Actions checks for malformed inputs, policy failures, and adversarial artifact fixtures; documented reproducible evidence and remaining security limits.

Business & Compliance Lead, AEROSEC (academic externship) Aug. 2025 - Dec. 2025

Honeywell Aerospace Technologies × Arizona State University *Tempe, AZ, USA*

- Developed compliance and commercialization analysis for a proposed airline PSS security layer, including a **$120K first-year commercialization scenario** with a five-year financial model presented to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader Jan. 2025 - Oct. 2025

Ira A. Fulton Schools of Engineering, Arizona State University *Mesa, AZ*

- Evaluated approximately **85 undergraduate web-development submissions per semester** and **52 graduate cybersecurity assignments**, giving evidence-based feedback on security controls, policy, compliance, risk analysis, and implementation quality.

Projects

Project periods are self-reported; CI figures are dated 2026 snapshots (UTC). Fixture results cover committed samples.

MCP Agent Security Gateway | *Python, FastAPI, MCP/JSON-RPC 2.0, Elastic Security* Oct. 2025 - Sep. 2026

- Implemented tool/server allowlists, egress policies, injection and PII inspection, rate limits, and hash-chained audit logging across supported gateway paths.
- Maintained **69 injection-pattern entries** and **9 Elastic rule records**, with **21 core SIEM tests**; controls vary by integration path.
- Oct. 10 Python 3.12 CI: **782 passed** at **81.93% statement coverage** in the same run.

AWS Agent Identity Guard | *Python, AWS IAM, SARIF 2.1.0* Apr. 2025 - Sep. 2025

- Implemented **24 offline policy/trust rules** plus a permission-boundary check for optional role metadata, covering wildcard grants, PassRole, AssumeRole, and escalation risks.
- Produced text, JSON, and SARIF 2.1.0 findings for policy review and GitHub Code Scanning; static analysis does not establish complete effective permissions.
- Oct. 10 CI: **247 collected, 244 passed, 3 skipped**; skips are disabled separate-rule-API tests.

HF Model Provenance Scanner | *Python, SafeTensors, GGUF, ONNX, Keras* Nov. 2024 - Mar. 2025

- Built static, non-executing inspection for SafeTensors, GGUF, ONNX, Keras, and pickle-derived artifacts, plus Python-loader and dependency checks.
- Detected **33/33 committed adversarial fixtures**, with **0 actionable findings on 4 committed benign samples**; these are bounded fixture results.
- Oct. 9 Python 3.12 CI: **260 passed, 1 skipped** with **75.67% statement coverage**.

Education

Arizona State University Tempe, AZ, USA

*M.S., Information Technology (Security), GPA: **3.87/4.00*** *Aug. 2024 - May 2026*

- **Academic Participation:** Student participant, ASU, 2025; Generative AI learning assistant for personalized education.

M. S. Ramaiah University of Applied Sciences Bengaluru, KA, India

*B.Tech., Computer Science & Engineering, CGPA: **8.44/10*** *Aug. 2019 - Aug. 2023*

Technical Skills

AI & Agent Security: MCP/JSON-RPC 2.0, Tool/Function-Call Security, Prompt Injection, LLM Red Teaming, Model Supply-Chain Security

Identity & Application Security: Cloud IAM Security, AWS IAM, Least Privilege, Trust Policies, Permission Boundaries, Capability Authorization, Threat Modeling, API Security

Security Engineering & DevSecOps: Python, FastAPI, pytest, GitHub Actions, CI/CD Security, SARIF 2.1.0, GitHub Code Scanning, Docker

Detection & Observability: Elastic Security, ECS, SIEM, Prometheus, Tamper-Evident Audit Logging, MITRE ATLAS

Certifications

AWS Academy Graduate - Cloud Architecting (Apr. 2025) | AWS Academy Graduate - Cloud Security Foundations (Nov. 2025)